AME 323: Project 1 hints

Taylor-Maccoll solver

1 Using **ode45**

One straightforward way to solve the Taylor-Maccoll equation numerically is to use the Matlab® built-in function `ode45`. This function is fast and accurate when solving nonstiff initial value problems.

In order to format the equation in a way that `ode45` can solve it, we need to convert any higher-order ODEs into a system of first-order ODEs. One example of a second-order, nonlinear ODE given in the Matlab® reference literature is the van der Pol equation:

$$\phi'' - \mu(1 - \phi^2)\phi' + \phi = 0. \quad (1)$$

In order to rewrite this as a system of first-order ODEs to be solved numerically, we introduce a change of variables:

$$y_1 = \phi, \quad (2a)$$
$$y_2 = \phi' = y_1'. \quad (2b)$$

We then substitute Equation 2 into Equation 1 to arrive at the following:

$$y_1' = y_2, \quad (3a)$$
$$y_2' = \mu(1 - y_1^2)y_2 - y_1. \quad (3b)$$

This is now a system of first-order ODEs that can be coded into a Matlab® function and solved by `ode45`. The online literature shows how to do this, however, it may be more clear to write the function given online as follows.

```
function dydt = vdp1(t,y,mu)
%VDP1  The van der Pol ODEs

dydt = zeros(2,1);

dydt(1) = y(2);
dydt(2) = mu*(1-y(1)^2)*y(2)-y(1);

end
```

Here, μ is passed to the function as a parameter from outside of the function (in the main script). This same concept can be used with the Taylor-Maccoll equation, which also has a parameter γ to pass from the main script.

For more information on all of the ODE solvers available in Matlab® (which all use the same syntax), see: Choose an ODE solver.

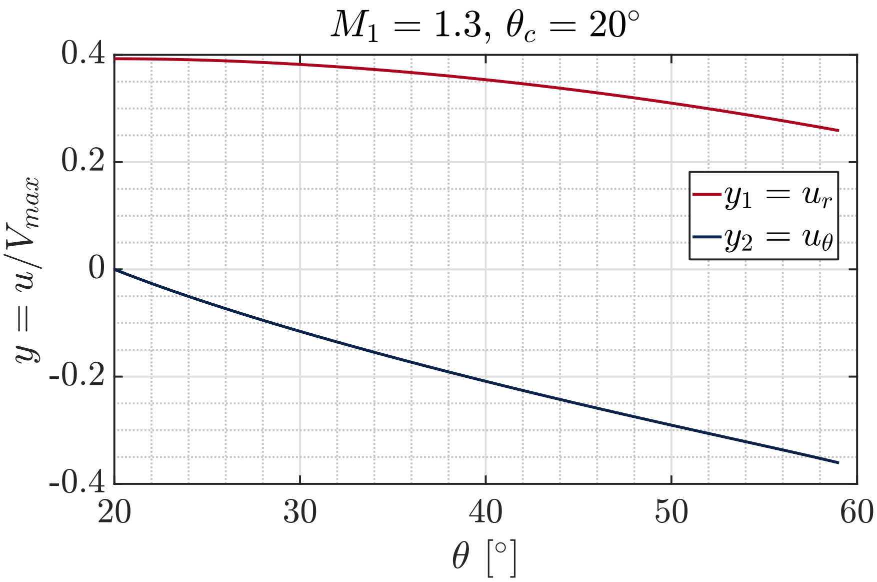

Figure 1: Two example solutions for $M_1 = 1.3$ and $\theta_c = 20°$.

2 Behavior near the extremes for each Mach number

Each Mach number M_1 has two extreme cases, zero deflection angle and detachment (maximum deflection angle). The behavior of your solver is likely to be problematic near these two extremes. Initial guesses and setting limits on β will be important here.

Near the zero deflection angle ($\theta_c = 0$) limit, we already know what value β should approach based on the case of the infinitely weak oblique shock. That example applies whether the flow is over a wedge or a cone.

Near the detachment limit, problems can also arise due to the rapid change in β and the fact that a strong shock solution also exists nearby. Being careful with initial guesses, taking small step sizes, and trying other solvers are several options for overcoming this issue. It is likely that computation times will be longer in these areas.

3 Behavior of solutions

Solutions to the Taylor-Maccoll equation should not oscillate. One example solution is shown in Figure 1.

